MAT 110 Differential Calculus & Coordinate Geometry

Week 3

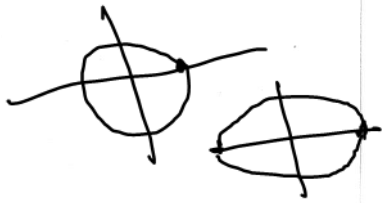
- → Parametric differentiation
- → Differentiation with the help of introducing 'ln'
- → Differentiation of inverse function
- → Leibnitz's theorem
- → Gradient & Tangent line.

Parametric differentiation:

Ex $x = x(t)$, $y = y(t)$ OR $x = x(\theta)$, $y = y(\theta)$

① Consider
$x = 2t$, $y = t^2$ } → pair of these together called parametric eqn
→ t = parameter

② $x = \sin\theta + 2$, $y = \cos\theta - 3$ } → parametric eqn
θ = parameter



If a function is expressed in terms of some parameter, in other words if the equation of the function is given parametrically: considering eqn of an ellipse

$x = a\cos t$, $y = b\sin t$, $0 \le t \le 2\pi$

The chain Rule can be used to find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} \longrightarrow \frac{dy/dt}{dx/dt}$$ → while $\frac{dx}{dt} \neq 0$

Examples can be found in Week 2 PDf **pg 16, 17 Ex 5, 6, 7**

Differentiation with the help of Introducing 'ln'

The function can be differentiated by taking 'ln' if:

- The given function is:
  - → product of some different functions
  - → quotient of some different functions

- power of a function is another function
  - ↳ see examples in Week 2 PDF

Ex 4 pg 15, Ex 5 pg 20

product of some different functions:

$y = pqr \qquad p = p(x),\ q = q(x),\ r(x)$

$\ln y = \ln(pqr)$

$= \ln p + \ln q + \ln r$

differentiate w.r.t. x on both sides

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{p}\frac{dp}{dx} + \frac{1}{q}\cdot\frac{dq}{dx} + \frac{1}{r}\frac{dr}{dx}$$

$$\frac{dy}{dx} = y\left[\frac{1}{p}\cdot\frac{dp}{dx} + \frac{1}{q}\frac{dq}{dx} + \frac{1}{r}\cdot\frac{dr}{dx}\right]$$

$$\frac{dy}{dx} = pqr\left[\frac{1}{p}\cdot\frac{dp}{dx} + \frac{1}{q}\cdot\frac{dq}{dx} + \frac{1}{r}\frac{dr}{dx}\right]$$

quotient of some different functions:

$$y = \frac{pq}{r}, \quad p = p(x),\ q = q(x),\ r = r(x)$$

$$\ln y = \ln\left(\frac{pq}{r}\right)$$

$$\ln y = \ln p + \ln q - \ln r$$

differentiate w.r.to x on both sides:

$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{p}\cdot\frac{dp}{dx} + \frac{1}{q}\cdot\frac{dq}{dx} - \frac{1}{r}\cdot\frac{dr}{dx}$$

$$\frac{dy}{dx} = y\left[\frac{1}{p}\cdot\frac{dp}{dx} + \frac{1}{q}\cdot\frac{dq}{dx} - \frac{1}{r}\cdot\frac{dr}{dx}\right]$$

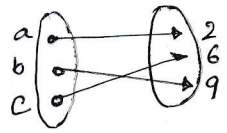
$$\frac{dy}{dx} = \frac{pq}{r}\left[\frac{1}{p}\cdot\frac{dp}{dx} + \frac{1}{q}\cdot\frac{dq}{dx} - \frac{1}{r}\cdot\frac{dr}{dx}\right]$$

Differentiation of Inverse function:

See Ex 2, 3 pg 14 of week 2 pdf

- Every function does not have its inverse
- In order to have an inverse, the function must be one-to-one.

A one-to-one function is a function of which the answers never repeat. For example the function $f(x) = x+1$ is a one-to-one function because it produces a different answer for every input.



(2) Inverse function & chain rule:

Find the derivative of $f(x) = \ln(\sin^{-1}(x^2-3x))$.

$$f'(x) = \frac{1}{\sin^{-1}(x^2-3x)} \frac{d}{dx} \sin^{-1}(x^2-3x)$$

$$= \frac{1}{\sin^{-1}(x^2-3x)} \cdot \frac{1}{\sqrt{1-(x^2-3x)^2}} \cdot \frac{d}{dx}(x^2-3x)$$

$$= \frac{1}{\sin^{-1}(x^2-3x)} \cdot \frac{1}{\sqrt{1-(x^2-3x)^2}} \cdot (2x-3)$$

$$= \frac{2x-3}{\sin^{-1}(x^2-3x)\sqrt{1-(x^2-3x)^2}}.$$

(3) Inverse Function & Implicit Differentiation

Find $\frac{dy}{dx}$ for $\cos^{-1}(xy) = x^2$

$$\frac{d}{dx}(\cos^{-1}(xy)) = \frac{d}{dx}(x^2)$$ [derive both sides w.r.t. x]

$$\frac{-1}{\sqrt{1-(xy)^2}} \frac{d}{dx}(xy) = 2x$$

$$\frac{-\left(1 \cdot y + x \cdot \frac{dy}{dx}\right)}{\sqrt{1-(xy)^2}} = 2x$$

$$\frac{-y - x\frac{dy}{dx}}{\sqrt{1-(xy)^2}} = 2x$$

$$-y - x\frac{dy}{dx} = 2x\sqrt{1-(xy)^2}$$

$$-x\frac{dy}{dx} = 2x\sqrt{1-(xy)^2} + y$$

$$\frac{dy}{dx} = -\frac{1}{x}\left[2x\sqrt{1-(xy)^2} + y\right]$$

$$= -2\sqrt{1-x^2y^2} - \frac{y}{x}$$